



Mortality and in-patient outcomes in pheochromocytoma patients with hypertensive emergency in the United States: A propensity score matched analysis

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ABSTRACT

Introduction: Pheochromocytoma is a rare cause of hypertensive emergency. The objective of this analysis was to compare the clinical characteristics, comorbidities, and in-hospital outcomes of patients admitted with hypertensive emergencies with and without co-existing pheochromocytoma.

Methods: A retrospective analysis of the National Inpatient Sample (NIS) Database from 2016 to 2020 was conducted, encompassing 640,395 patients hospitalized for hypertensive emergencies, including 2535 patients diagnosed with pheochromocytoma. We compared demographics, comorbidities, in-hospital outcomes and resource utilization metrics in patients with and without pheochromocytoma. Propensity-score matching was utilized to account for potential confounders and risk of complications was compared.

Results: Among the pheochromocytoma cohort (51.9% female), a significant portion (35.7%) were under 50 years of age, with the majority being Caucasian (47.9%). Comorbid conditions such as obesity, diabetes, and smoking were prevalent, with notable differences in cancer (7.5% vs. 2.3%, $p < 0.001$) and peripheral vascular disease (17% vs. 8.2%, $p < 0.001$) rates compared to the non-pheochromocytoma cohort. Pheochromocytoma patients had a longer hospital stay (7.5 vs. 6 days, $p = 0.002$) and higher odds of acute kidney injury (AKI) (1.54, 1.18-2, $p = 0.001$) but lower odds of requiring hemodialysis (0.52, 0.32-0.79, $p < 0.001$) or experiencing major cardiovascular events (0.5, 0.36-0.69, $p < 0.001$). No significant difference in inflation-adjusted hospitalization costs was found between the groups.

Conclusions: Patients with hypertensive emergencies and pheochromocytoma had a higher incidence of AKI, certain comorbidities (cancer, peripheral vascular disease), and more complex hospital courses suggested by longer length of stay. However, the overall cost of hospitalization did not significantly differ between the two cohorts.

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Table 1

Characteristics of patients in hypertensive emergency with and without pheochromocytoma.

Characteristics	Hypertensive Emergency & Pheochromocytoma-		Hypertensive Emergency & Pheochromocytoma+		P-value
	N	%	N	%	
N = 640,930	638395	99.6	2535.0	0.4	
GENDER (%)	N	%	N	%	<0.001
Female	279220	43.7	1315.0	51.7	
Male	359175	56.3	1220.0	48.1	
MEAN AGE YEARS (SD)	Mean	SD	Mean	SD	
Female	53.56	11.6	52.4	11.9	
Male	52.87	10.9	50.5	10.9	
AGE GROUPS (%)	N	%	N	%	<0.001
18-29	21725	3.4	110.0	4.3	
30-49	192815	30.2	905.0	35.7	
50-69	423855	66.4	1520.0	60.0	
RACE (%)	N	%	N	%	<0.001
Asian or Pacific Islander	19030	3.0	90.0	3.6	
Black	282490	44.3	975.0	38.5	
Hispanic	79790	12.5	180.0	7.1	
Native American	3445	0.5	15.0	0.6	
Other	20275	3.2	60.0	2.4	
White	233365	36.6	1215.0	47.9	
MEDIAN HOUSEHOLD INCOME (\$)	N	%	N	%	<0.001
<=49,999	283435	44.4	1065.0	42.0	
50K-64,999	161050	25.2	585.0	23.1	
65K-85,999	120095	18.8	500.0	19.7	
>=86k	73815	11.6	385.0	15.2	
INSURANCE STATUS (%)	N	%	N	%	<0.001
Medicaid	165035	25.9	710.0	28.0	
Medicare	212870	33.3	645.0	25.4	
No charge	6650	1.0	20.0	0.8	
Other	17965	2.8	35.0	1.4	
Private Insurance	160180	25.1	795.0	31.4	
Self-pay	75695	11.9	330.0	13.0	
HOSPITAL DIVISION (%)	N	%	N	%	<0.001
East North Central	90745	14.2	460.0	18.2	
East South Central	52555	8.2	180.0	7.1	
Middle Atlantic	70190	11.0	315.0	12.4	
Mountain	31585	5.0	120.0	4.7	
New England	16745	2.6	110.0	4.3	
Pacific	74000	11.6	280.0	11.1	
South Atlantic	173775	27.2	625.0	24.7	
West North Central	29875	4.7	125.0	4.9	
West South Central	98925	15.5	320.0	12.6	
HOSPITAL BEDSIZE (%)	N	%	N	%	<0.001
Large	336145	52.7	1420.0	56.0	
Medium	187615	29.4	675.0	26.6	
Small	114635	18.0	440.0	17.4	
HOSPITAL TEACHING STATUS (%)	N	%	N	%	<0.001
Rural	32050	5.0	130.0	5.1	
Urban nonteaching	105265	16.5	305.0	12.0	
Urban teaching	501080	78.5	2100.0	82.8	
COMORBIDITIES (%)	N	%	N	%	
Diabetes	282465	44.3	975.0	38.5	<0.001
Cancer	14545	2.3	190.0	7.5	<0.001
Obesity	173980	27.3	775.0	30.6	<0.001
Drug Abuse	68015	10.7	170.0	6.7	<0.001
Smoking	284765	44.6	1270.0	50.1	<0.001
Alcohol	48025	7.5	130.0	5.1	<0.001
Chronic Pulmonary Disease	129895	20.4	415.0	16.4	<0.001
Peripheral Vascular Disease	52430	8.2	430.0	17.0	<0.001
Hypothyroidism	44305	6.9	230.0	9.1	<0.001
Autoimmune	21775	3.4	90.0	3.6	0.699
Depression	69695	10.9	400.0	15.8	<0.001
AIDS	8790	1.4	15.0	0.6	<0.001
Dementia	10255	1.6	40.0	1.6	0.909

N = N = Number, % = Percentage, SD= standard deviation, \$ (US Dollar).

Introduction

Hypertensive emergency represents a critical medical condition characterized by severe hypertension leading to organ damage, including myocardial infarction, pulmonary edema, stroke, and acute kidney injury. This condition, though rare with an estimated incidence of 0.5% among emergency department (ED) visits, carries a significant mortality risk—ranging from 5–13% during hospitalization to 26–39% within the first 30 days post-presentation.^{1,2} Recent epidemiological data indicate a rising trend in hypertensive emergency presentations, with ED visits doubling from 2006 to 2013.²

Pheochromocytoma, a rare tumor causing hypertension, has an incidence rate of approximately 0.6 cases per 100,000 person-years and represents a mere 0.1% of all hypertension cases in the United States.^{3,4} Distinguishing pheochromocytoma from other causes of hypertensive emergencies is critical due to its unique management approach. Unlike the general treatment of hypertensive emergencies, which involves titrating antihypertensive medications and transitioning to more potent oral regimens, pheochromocytoma requires surgical intervention to remove the catecholamine-secreting tumor, often resulting in a curative outcome.^{5–9}

Despite the clear differentiation in treatment approaches between hypertensive emergencies caused by pheochromocytoma and other etiologies, there is a scarcity of high-quality data directly comparing inpatient outcomes of these two patient populations. Addressing this gap, our study leverages the National Inpatient Sample (NIS) database to explore the clinical characteristics and inpatient outcomes of patients admitted with hypertensive emergencies, including those diagnosed with pheochromocytoma. A secondary objective of our study is to assess the impact of pheochromocytoma on the outcomes of patients admitted for hypertensive emergencies.

Table 2
In-hospital outcomes & complications.

Characteristics	Hypertensive Emergency & Pheochromocytoma-		Hypertensive Emergency & Pheochromocytoma+		P-value
	N	%	N	%	
N = 640,930	638395	99.60	2535	0.40	–
Disposition (%)	N	%	N	%	<0.001
Against medical advice	28280	4.43	120	4.73	
Died in Hospital	19565	3.06	50	1.97	
Discharged alive unknown destination	15	0.00	0	0.00	
Home health care	64265	10.07	250	9.86	
Routine	426005	66.73	1795	70.81	
Transfer other	84005	13.16	250	9.86	
Transfer to short-term hospital	16260	2.55	70	2.76	
Complications (%)	N	%	N	%	
Vasopressor use	5500	0.86	45	1.78	0.077
	Adjusted OR 1.82 (CI 0.94-3.55)				
AKI	217085	34.00	1070	42.21	<0.001
	Adjusted OR 1.45 (CI 1.21-1.74)				
Invasive Mechanical Ventilation	76120	11.92	265	10.45	0.391
	Adjusted OR 0.88 (CI 0.66-1.18)				
Non-Invasive Mechanical Ventilation	45800	7.17	120	4.73	0.113
	Adjusted OR 0.72 (CI 0.47-1.08)				
Hemodialysis	106985	16.76	175	6.90	<0.001
	Adjusted OR 0.42 (CI 0.3-0.6)				
Mechanical Circulatory Support	1940	0.30	20	0.79	0.064
	Adjusted OR 2.56 (CI 0.95-6.93)				
AMI+Sudden Cardiac Arrest+CVA	185755	29.1	455	17.95	<0.001
	Adjusted OR 0.52 (CI 0.42-0.66)				
Acute Liver Failure	2950	0.46	25	0.99	0.130
	Adjusted OR 1.99 (CI 0.82-4.83)				
Encephalopathy	37360	5.85	150	5.92	0.795
	Adjusted OR 0.95 (CI 0.66-1.38)				
Mean Inflation-Adjusted Cost (\$)	\$19,017.88		\$22,381.83		0.152
	Adjusted Inflation-adjusted cost* = \$1,814.87 higher for Pheochromocytoma+				
Mean length of stay (days)	6.03		7.47		0.002
	Adjusted length of stay* = 1.20 days higher for Pheochromocytoma+				
In hospital mortality (N = 19,615)	19565	3.06	50	1.97	0.146
	Adjusted OR 0.63 (CI 0.33-1.18)				

N = Number, % = Percentage, CI = Confidence Interval, AKI = Acute kidney injury, AMI = Acute myocardial infarction, CVA = cerebrovascular accident, \$ (US Dollar). * Adjusted for Age, Hospital bedsize, Race, Gender, Hospital location, Hospital teaching status, Hospital region, Median Household Income, Expected primary payer (insurance status), Elixhauser comorbidities. * Adjusted for Age, Hosp Bedsize, Race, Gender, Hospital location, Hospital teaching status, Hospital region, Median Household Income, Expected primary payer (insurance status), ELIXHAUSER COMORBIDITIES.

Methods

Data source

Data were obtained from the United States National Inpatient Sample (NIS) database, which is the largest publicly available inpatient database compiling all pay sources in the United States. Developed with the support of the Agency of Healthcare Research and Quality (AHRQ) and the Healthcare Cost and Utilization Project (HCUP), the NIS aims to produce regional and national estimates of inpatient utilization, access, cost and outcomes. This dataset contains information from approximately 7 million hospital stays each year.¹⁰ Given that this study utilized only de-identified data, an institutional review board approval was not required.¹¹

Study design and statistical analysis

This was a retrospective cohort study analyzing NIS database records of all patients admitted with a primary diagnosis of hypertensive emergency over a 5-year period between January 1, 2016 and December 31, 2020. The identification of patients with a primary diagnosis of hypertensive emergency and secondary diagnosis of pheochromocytoma were done using International Classification of Diseases, 10th Revision, Clinical Modification (ICD-10) codes (Supplemental Table 1). Patients with age <18 and incomplete data in the hospitalization records were excluded.

For statistical analyses, we first performed test for independence for the two cohorts against the categorical variables (gender, race, disposition, age group etc.), p-values for these tests are reported in Tables 1, 2. Small p-values imply that there exists a relationship between the categorical variable and the two cohorts. For the continuous variables, length of stay and mean inflation-adjusted total charge, we developed a multivariate linear regression model with the hospital-related variables, patients related data, and the Elixhauser comorbidities as predictors. Next, logistic regression models were built to report the odds ratios for the two cohorts for the comorbidities such as vasopressor use, acute kidney injury, and hemodialysis, and 95% confidence intervals are also reported for the odds ratios. Lastly, a Mann-Kendall test was performed to test presence of trend in mortality over the 5-year period. A propensity matched sample was derived from both the cohorts and the analyses repeated on the matched cohorts. SAS was used in data curation

Table 3

In-hospital outcomes & complications: propensity matched.

Characteristics	Hypertensive Emergency & Pheochromocytoma-		Hypertensive Emergency & Pheochromocytoma+		P-value
	N	%	N	%	
N = 5,070	2535	50.00	2535	50.00	–
Disposition (%)	N	%	N	%	<0.001
Against medical advice	95	3.75	520	3.88	
Died in Hospital	75	2.96	470	3.50	
Home health care	255	10.06	1125	8.39	
Routine	1710	67.46	9565	71.30	
Transfer other	350	13.81	1075	8.01	
Transfer to short-term hospital	50	1.97	660	4.92	
COMPLICATIONS (%)	N	%	N	%	
Vasopressor use	25	0.99	45	1.78	0.266
	Adjusted OR 1.87 (CI 0.62-5.66)				
AKI	820	32.35	1070	42.21	0.001
	Adjusted OR 1.54 (CI 1.18-2.0)				
Invasive Mechanical Ventilation	280	11.05	265	10.45	0.570
	Adjusted OR 0.89 (CI 0.59-1.34)				
Non-Invasive Mechanical Ventilation	175	6.9	120	4.73	0.116
	Adjusted OR 0.65 (CI 0.38-1.11)				
Hemodialysis	350	13.81	175	6.90	0.003
	Adjusted OR 0.5 (CI 0.32-0.79)				
AMI+Sudden Cardiac Arrest+CVA	715	28.21	455	17.95	<0.001
	Adjusted OR 0.5 (CI 0.36-				
Acute Liver Failure	15	0.59	25	0.99	0.482
	Adjusted OR 1.67 (CI 0.4-7.04)				
Encephalopathy	140	5.52	150	5.92	0.687
	Adjusted OR 1.12 (CI 0.65-1.92)				
Mean Inflation-Adjusted Cost (\$)	\$18,988.27		\$22,381.83		0.117
	Adjusted Inflation-adjusted cost* = \$2,736.46 higher for Pheochromocytoma+				
Mean length of stay (days)	5.79		7.47		0.001
	Adjusted length of stay* = 1.60 days higher for Pheochromocytoma+				
In hospital mortality (N = 125)	75	2.96	50	1.97	0.279

N = Number, % = Percentage, CI = Confidence Interval, AKI = Acute kidney injury, AMI = Acute myocardial infarction, CVA = cerebrovascular accident, \$ (US Dollar). * Adjusted for Age, Hospital bedsize, Race, Gender, Hospital location, Hospital teaching status, Hospital region, Median Household Income, Expected primary payer (insurance status), Elixhauser comorbidities. * Adjusted for Age, Hosp Bedsize, Race, Gender, Hospital location, Hospital teaching status, Hospital region, Median Household Income, Expected primary payer (insurance status), ELIXHAUSER COMORBIDITIES.

whereas the statistical analyses required the use of R and Python.

Results

Demographics and baseline comorbidities

Our study sample included 640,395 patients (43.8% female) hospitalized for hypertensive emergency. Among these 2,535 (51.9% female) patients with hypertensive emergency and pheochromocytoma, the majority (51.9%) were female. Around 1/3rd (35.7%) patients were under 50 years of age. The majority of patients were Caucasian (47.9%), followed by African American (38.5%). Most patients had household incomes below 50,000 USD (42%). Majority of patients were treated at large hospitals (56%) and urban teaching hospitals (82.8%).

The most common comorbidities in both pheochromocytoma and non-pheochromocytoma cohorts included smoking (50.1% vs. 44.6%), diabetes (38.5% vs. 44.3%), obesity (30.6% vs. 27.3%), chronic pulmonary diseases (16.4% vs. 20.4%), and depression (15.8% vs. 10.9%). The baseline characteristics and comorbidity burden of the pheochromocytoma and non-pheochromocytoma cohorts while statistically significant, demonstrated only small variances overall. However, notable exceptions were observed in gender distribution (Female: 51.9% vs. 43.7%, $p < 0.001$) and certain comorbidities including cancer (7.5% vs. 2.3%, $p < 0.001$) and peripheral vascular disease (17 vs. 8.2%, $p < 0.001$), which were more prevalent in the pheochromocytoma group (Table 1).

In-Hospital outcomes

The majority of patients in pheochromocytoma and non-pheochromocytoma groups were discharged to routine care (70.8% vs. 66.7%, $p < 0.001$). A small proportion of patients died during hospitalization (1.97% in pheochromocytoma cohort vs. 3.06% in non-pheochromocytoma cohort) (Table 2). Similar findings were noted in propensity matched analysis with majority of patients discharged to routine care (71.3% vs. 67.5%) and only a small fraction of patients died during in-hospital stay (3.5% in pheochromocytoma cohort vs. 3% in non-pheochromocytoma cohort). A small proportion of patients died during hospitalization (3.5% in pheochromocytoma cohort vs. 3% in non-pheochromocytoma cohort) (Table 2). Similar to baseline characteristics, although in-patient outcomes of pheochromocytoma and non-pheochromocytoma cohorts were statistically significant but demonstrated only small variances overall.

In-hospital complications

The pheochromocytoma cohort had higher odds of having acute kidney injury (AKI) (OR 1.4, CI 1.2-1.7, $p < 0.001$), but a significantly lower odds of having acute myocardial infarction, sudden cardiac death, or cerebrovascular accident (OR 0.5, CI 0.4-0.7, $p < 0.001$), and needing hemodialysis (OR 0.4, CI 0.3-0.6, $p < 0.001$). There was no statistically significant difference between two cohorts in requiring invasive mechanical ventilation (OR 0.88, CI 0.66-1.18, $p = 0.391$), non-invasive mechanical ventilation (OR 0.72, CI 0.47-1.08, $p = 0.113$), acute liver failure (OR 1.99, CI 0.82-1.38, $p = 0.13$), and encephalopathy (OR 0.95, CI 0.66-1.38, $p = 0.795$). Similar findings were noted in the propensity-matched cohorts (Table 3).

The mean inflation-adjusted cost of hospitalization of pheochromocytoma and non-pheochromocytoma cohort were \$22,381.83 and \$19,017.88 (\$1,814.87 higher cost in pheochromocytoma cohort, $p = 0.002$). However, no statistically significant difference was noted in hospitalization cost based on propensity score match data (\$22,381.83 vs. \$18,988.27, $p = 0.15$). Pheochromocytoma cohort had a statistically significant longer length of stay (LOS) in the hospital (7.5 vs. 6 days, $p = 0.002$) (Table 2). Similar findings were noted in the propensity-matched cohorts (7.5 vs. 5.8 days, $p = 0.001$) (Table 3).

Discussion

In this nationally representative retrospective analysis of patients admitted with hypertensive emergency using NIS Database from 2016 to 2020, we analyzed 2,535 patients (51.9% female) with pheochromocytoma. Key findings of our study included, a) around 1/3rd (35.7%) patients were under 50 years of age, b) only small variances in baseline characteristics, comorbidity burden and disposition between patients with and without pheochromocytoma presenting with hypertensive emergency with few notable exceptions, c) Compared to control cohort, Pheochromocytoma cohort had longer length of stay (LOS) in the hospital (7.5 vs. 6 days, $p = 0.002$). However, no statistically significant difference in hospitalization costs was observed between the two groups.

Baseline patient-level characteristics

It is estimated that incidence of pheochromocytoma ranges from 0.005% to 0.1% in general population and 0.1 to 0.6% in adult hypertensive population.^{12,13} Typically the prevalence of pheochromocytoma peaks in 3rd and 5th decades of life.¹⁴ Pheochromocytoma crisis is a life threatening emergency characterized by hemodynamic instability (severe hypertension or hypotension) with multi-organ failure.^{15,16} In a review of 200 case reports on pheochromocytoma crisis, the mean age at presentation was 43.8 ± 15.5 years, with a predominance of female patients (55.5%). Our study reported similar findings, with a slight increase in mean age to 51.5 ± 11.4 years and a majority female representation (51.87%). This consistent gender distribution may suggest a gender-specific predisposition to pheochromocytoma crisis that require research. However, it is crucial to note that hypertensive crises represent just one clinical manifestation of pheochromocytoma; other presentations, including hypotensive crises, are also possible. Consequently, the

data from previous studies may not align perfectly with ours, highlighting a gap in the literature regarding hypertensive crises caused by pheochromocytoma. Our analysis also revealed significant racial and socioeconomic distribution trends within the pheochromocytoma cohort. A higher prevalence was noted among White patients, with a substantial number of cases treated in large, urban teaching hospitals. The intricacies involved in treating pheochromocytoma, which often require a comprehensive multidisciplinary approach and specialized perioperative care, may explain why these patients are predominantly managed in larger, urban academic medical centers.

Comorbidities

Our study has illuminated significant correlations between pheochromocytoma and specific comorbidities. Notably, cancer prevalence was markedly higher among pheochromocytoma patients (7.5% vs. 2.3%, $p < 0.001$). This finding is congruent with the established association between pheochromocytoma and genetic syndromes such as Multiple Endocrine Neoplasia Type 2 (MEN2), Von Hippel-Lindau Syndrome (VHL), Neurofibromatosis Type 1 (also known as Von Recklinghausen's Disease, VRD), and hereditary paraganglioma syndromes Types 1-5 (SDH complex).¹⁴

Additionally, the data indicates an increased incidence of peripheral vascular diseases (17% vs. 8.2%, $p < 0.001$) and depression (15.8% vs. 10.9%, $p < 0.001$) among pheochromocytoma patients. The significance of these associations remains to be fully understood but may point toward the interplay between psychosocial factors, lifestyle, and the pathophysiology of pheochromocytoma.

In-hospital outcomes

In terms of in-hospital outcomes, the epidemiology of pheochromocytoma is relatively understudied. A mortality rate of 13.8% was documented in a review of 200 case reports of pheochromocytoma crisis.¹⁴ This is a marked reduction from the approximately 45% mortality rate reported in an earlier review of 11 cases between 1988-2009.¹⁷ Such a significant decline in mortality is generally attributed to advancements in early diagnosis and surgical intervention. Support for this theory comes from the work of Scholten et al., who reported a 2% mortality rate in their review of 58 surgically treated patients from 1990-2013.¹⁸ Our findings align with this trend, indicating a mortality rate of 2% during hospitalization for pheochromocytoma patients. These findings further emphasize the importance of early detection and appropriate surgical management in improving survival outcomes for patients with pheochromocytoma.

Hypertensive emergency: comparing patients with & without pheochromocytoma

Our comparative analysis of hypertensive emergency patients, with and without pheochromocytoma, highlighted distinct profiles in comorbidities and clinical outcomes.

We noted a higher incidence of AKI in pheochromocytoma cohort (42.2% vs. 32.3%, OR 1.54, $P = 0.001$). This might be reflective of the hemodynamic instability and catecholamine excess associated with pheochromocytoma. Interestingly, patients in pheochromocytoma cohort showed a lesser propensity for myocardial infarction, sudden cardiac death, or cerebrovascular accidents or requiring HD. This finding could potentially be explained by the episodic nature of catecholamine secretion in pheochromocytoma, as opposed to the sustained hypertension seen in other etiologies.

In terms of hospitalization outcomes, the pheochromocytoma group, while at a lower risk for chronic cardiovascular events, experienced a longer stay in the hospital—a finding that persisted even after propensity score matching. The longer length of stay for pheochromocytoma patients is consistent with the anticipated complexity in managing such cases, often requiring a multidisciplinary approach including preoperative optimization, surgery, and postoperative care.¹⁹⁻²⁴

Surprisingly, our study found no significant difference in inflation-adjusted hospitalization costs between the two cohorts. This could suggest that the higher costs associated with the medical and surgical management of pheochromocytoma might be balanced by the costs incurred from managing acute cardiovascular events in the non-pheochromocytoma hypertensive emergencies.

This study has several limitations. As a retrospective analysis based on ICD-10 codes, this investigation was constrained in the data that it could incorporate. We did not have access to laboratory or imaging data, for instance, which precluded us from analyzing the proportion of patients presenting with hypertensive emergency who were screened for pheochromocytoma. Thus, it remains possible that pheochromocytoma was simply undetected in patients classified into the non-pheochromocytoma group, underscoring the extent to which the reliance on ICD-10 coding introduces the potential for misclassification bias. Furthermore, this study only includes outcomes up to hospital discharge and does not incorporate important post-discharge metrics such as readmission rates or 30-day or 5-year survival rates, preventing an interrogation of long-term morbidity and mortality in these patient populations. Future studies are warranted to prospectively corroborate these findings and further characterize these high-risk diagnoses, with the ultimate goal of improving the prevention, risk stratification and treatment of hypertensive emergency.

Conclusions

In this retrospective analysis of the NIS database of patients admitted with hypertensive emergency with and without pheochromocytoma, we noted distinctive comorbidity profile and higher rates of AKI with longer hospital stay in pheochromocytoma cohort, without any significant difference in the overall hospitalization costs and inpatient mortality. Future longitudinal studies are needed to better understand the long-term impact of pheochromocytoma on patients with hypertensive emergencies, and to further refine

treatment guidelines and improve patient outcomes.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cpcardiol.2024.102578](https://doi.org/10.1016/j.cpcardiol.2024.102578).

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